

05_statistical_tests.R

Complete source code for reference. The original .R file must remain in GitHub for execution.

```
# Statistical tests supporting the analysis.
# Base R only: no external packages are required.

data <- read.csv("outputs/clean_advertising_campaigns.csv", stringsAsFactors = FALSE)
data$platform <- factor(data$platform)
data$campaign_type <- factor(data$campaign_type)
data$audience <- factor(data$audience)
train <- subset(data, split == "train")
model <- lm(revenue ~ spend + impressions + clicks + conversions + platform + campaign_type + audience, data=train)

platform_anova <- summary(aov(revenue ~ platform, data=train))[[1]]
anova_f <- platform_anova[1, "F value"]
anova_p <- platform_anova[1, "Pr(>F)"]
cor_test <- cor.test(train$spend, train$revenue, method="pearson")

e2 <- residuals(model)^2
fit <- fitted(model)
aux <- lm(e2 ~ fit)
bp_stat <- nrow(train) * summary(aux)$r.squared
bp_p <- pchisq(bp_stat, df=1, lower.tail=FALSE)
shapiro <- shapiro.test(residuals(model))

tests <- data.frame(
  test=c("Platform revenue ANOVA", "Pearson correlation: spend vs revenue", "Breusch-Pagan-style variance check", "Shapiro-Wilk test"),
  statistic=c(anova_f, unname(cor_test$estimate), bp_stat, unname(shapiro$statistic)),
  p_value=c(anova_p, cor_test$p.value, bp_p, shapiro$p.value)
)
write.csv(tests, "outputs/statistical_tests.csv", row.names=FALSE)
capture.output({
  print(summary(aov(revenue ~ platform, data=train)))
  print(cor_test)
  print(shapiro)
}, file="outputs/statistical_test_details.txt")
cat("Statistical testing completed.\n")
```